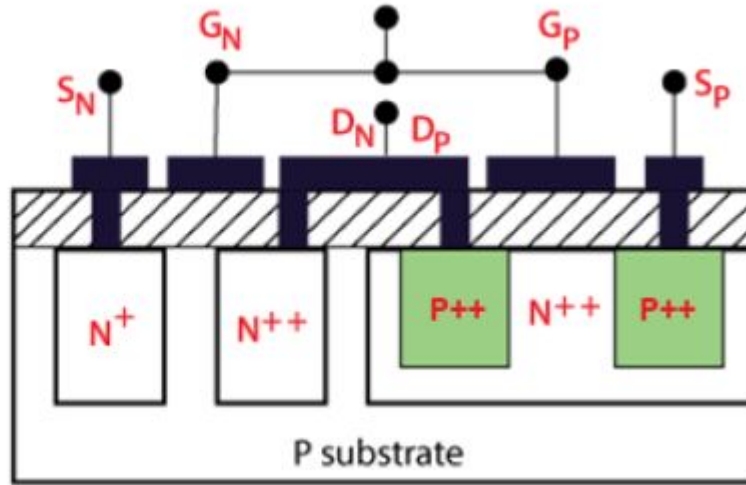


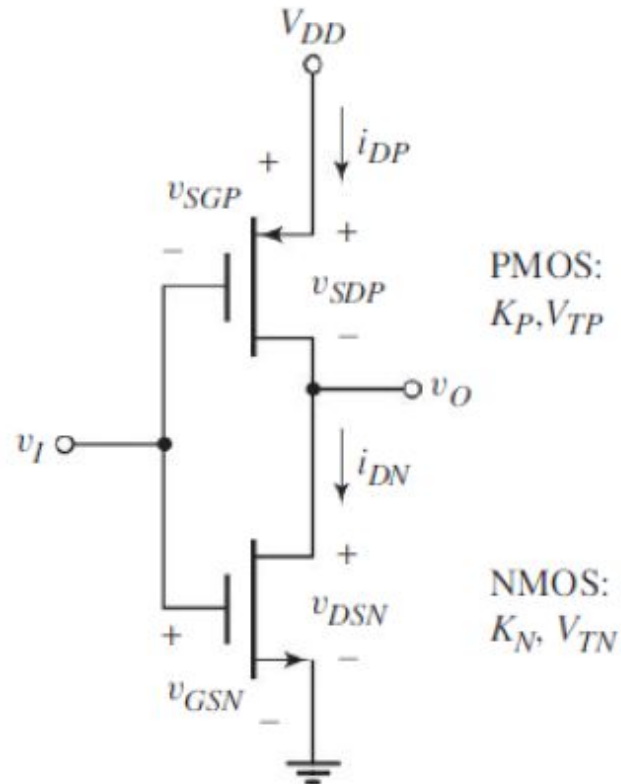
CSE350: Digital Electronics & Pulse Techniques

Lecture 14: CMOS Logic

CMOS Structure:



CMOS Inverter:



NMOS

Condition on V_{GS}	Condition on V_{DS}	Mode of Operation
$V_{GS} < V_{TN}$	ALL	Cut-off
$V_{GS} > V_{TN}$	$V_{DS} < V_{GS} - V_{TN}$	Linear
$V_{GS} > V_{TN}$	$V_{DS} > V_{GS} - V_{TN}$	Saturated

PMOS

Condition on V_{SP}	Condition on V_{SD}	Mode of Operation
$V_{SG} < -V_{TP}$	ALL	Cut-off
$V_{SG} > -V_{TP}$	$V_{SD} < V_{SG} + V_{TP}$	Linear
$V_{SG} > -V_{TP}$	$V_{SD} > V_{SG} + V_{TP}$	Saturated

Figure 16.25 CMOS inverter

VTC (Voltage Transfer Characteristics) of NMOS Inverter (with resistive load):

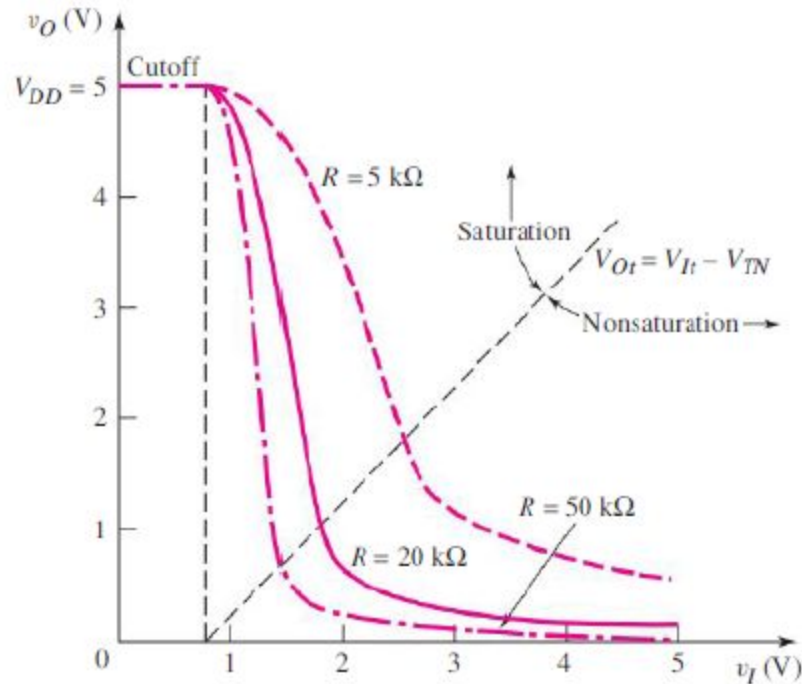
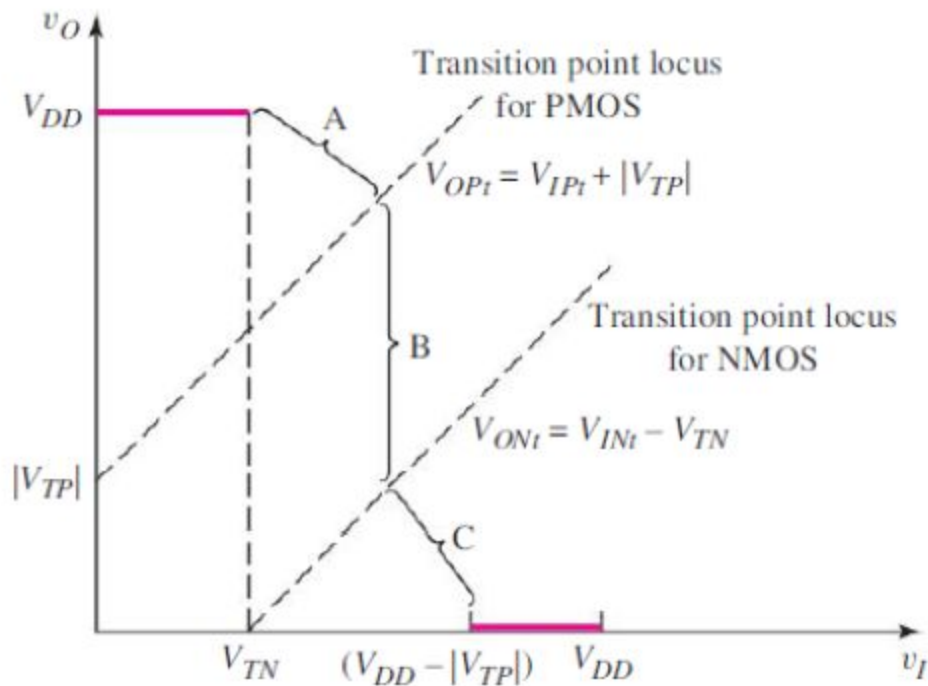
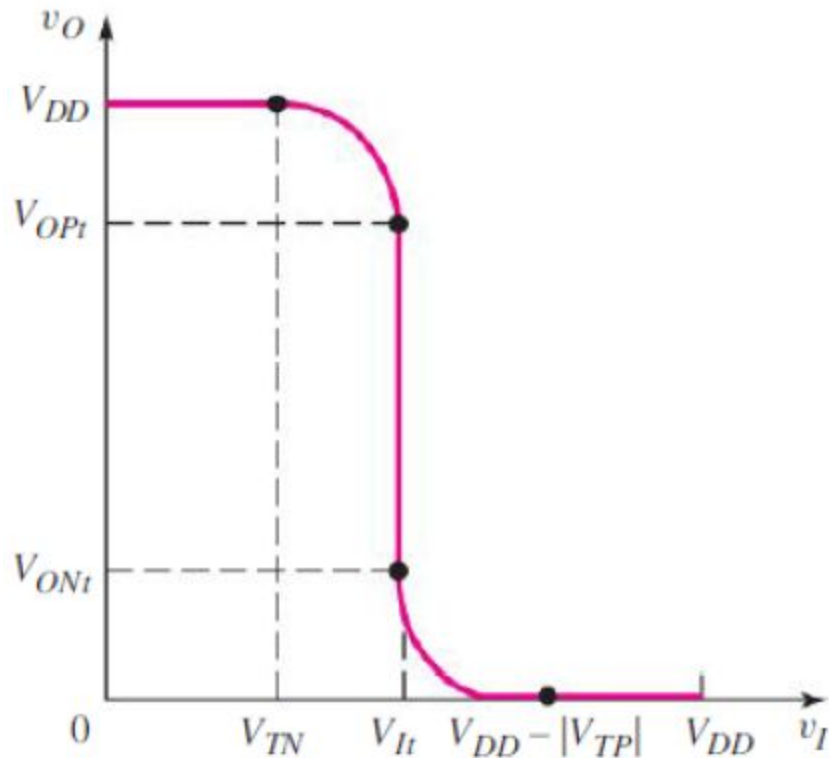


Figure 16.6 Voltage transfer characteristics, NMOS inverter with resistor load, for three resistor values

CMOS Inverter VTC:



Regions of operation:

1: $0 \leq V_I \leq V_{TN}$: **NMOS: Cutoff, PMOS: Triode**

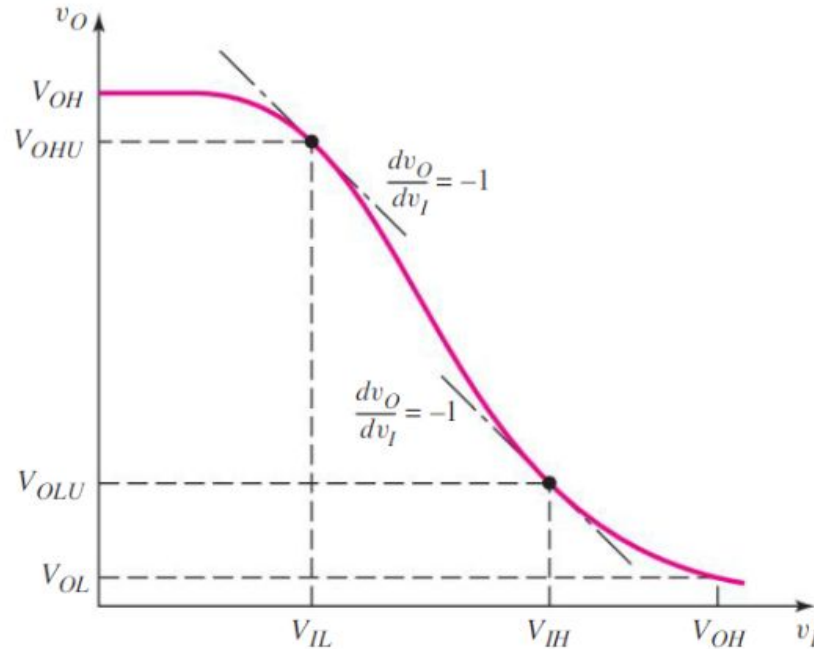
2: $V_{TN} \leq V_I \leq V_{IPt}$ = *transition voltage of PMOS*: **NMOS: Saturation, PMOS: Triode**

3: $V_{IPt} \leq V_I \leq V_{INt}$ = *transition voltage of NMOS*: **NMOS: Saturation, PMOS: Saturation**

4: $V_{INt} \leq V_I \leq V_{DD} - |V_{TP}|$: **NMOS: Triode, PMOS: Saturation**

5: $V_{DD} - |V_{TP}| \leq V_I \leq V_{DD}$: **NMOS: Triode, PMOS: Cutoff**

CMOS Inverter Noise Margin:



Low noise margin, $NM_L = V_{IL} - V_{OLU}$

High noise margin, $NM_H = V_{OHU} - V_{IH}$

Figure 16.35 Generalized inverter voltage curve and defined voltage limits V_{IL} and V_{IH}

Example 16.9: Determine the noise margin for CMOS Inverter: It is given,

$$K_p = K_n, V_{TN} = -V_{TP} = 0.4V$$

Solution: V_{OHU} & V_{IL} calculation:

From VTC of CMOS Inverter, we see that for this point, the NMOS is in saturation & PMOS is in triode mode. KCL gives,

$$I_{DP} = I_{DN}$$

$$\text{Or, } K_p [2(V_{SGP} + V_{TP})V_{SDP} - V_{SDP}^2] = K_n (V_{GSN} - V_{TN})^2$$

$$\text{Or, } 2(V_{SP} - V_{GP} + V_{TP})(V_{SP} - V_{DP}) - (V_{SP} - V_{DP})^2 = (V_{GN} - V_{SN} - V_{TN})^2$$

$$\text{Or, } 2(3.3 - V_I - 0.4)(3.3 - V_o) - (3.3 - V_o)^2 = (V_I - 0.4)^2$$

$$\text{Let, } V_I - 0.4 = X, 3.3 - V_o = Y. \text{ Or, } V_I = X + 0.4, V_o = 3.3 - Y$$

So, the previous KCL equation can be replaced by these new variables as,

$$2(2.5 - X)Y - Y^2 = X^2 \Rightarrow X^2 + Y^2 + 2XY - 5Y = 0$$

Now, $\frac{dY}{dX} = \frac{dY}{dV_o} \frac{dV_o}{dV_I} \frac{dV_I}{dX} = (-1)(-1)(1) = 1$

$$\left[\frac{dY}{dV_o} = \frac{d(3.3 - V_o)}{dV_o} = -1, \frac{dV_I}{dX} = \frac{d(X + 0.4)}{dX} = 1 \right]$$

Differentiating the KCL (replaced) equation to replace $\frac{dY}{dX} = 1$,

$$\begin{aligned} \frac{d}{dX} (X^2 + Y^2 + 2XY - 5Y) &= 0 \Rightarrow 2X + 2Y \frac{dY}{dX} - 5 \frac{dY}{dX} = 0 \Rightarrow 2X + 2Y - 5 = 0 \\ \Rightarrow Y &= \frac{5}{4} - X \end{aligned}$$

Replacing this in the original equation,

$$\begin{aligned} X^2 + \left(\frac{5}{4} - X\right)^2 + 2X\left(\frac{5}{4} - X\right) - 5\left(\frac{5}{4} - X\right) &= 0 \Rightarrow X = \frac{15}{16}, Y = \frac{5}{16} \Rightarrow V_{IL} = V_I = X + 0.4 \\ \Rightarrow V_{IL} &= 1.3375V, V_{OHU} = V_o = 3.3 - Y = 2.4875V \end{aligned}$$

Do similar analysis to find V_{IH} & V_{OLU} yourself.

Performing similar analysis, we get, $V_{IH} = 1.9625V$, $V_{OHU} = 0.3125V$

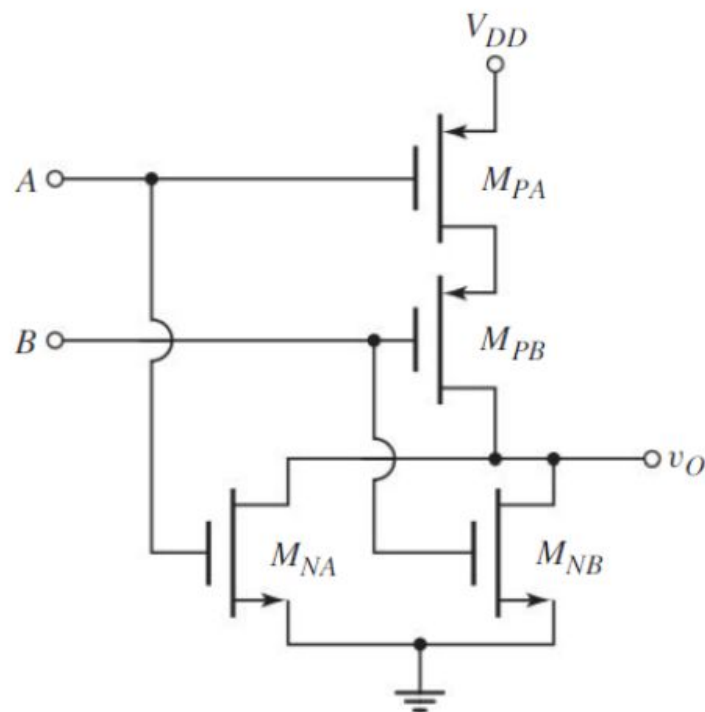
Thus, $NM = \min(NM_L, NM_H) = \min(V_{IL} - V_{OLU}, V_{OHU} - V_{IH}) = 1.025V$

Logic Circuits using CMOS:

Points to remember:

1. High input turns on/off the NMOS/PMOS, while the low voltage does the opposite.
2. Off MOSFET will result in open circuit, and on ones in short circuit between Drain & Source.
3. **Conduction complement:** If the intended logic function implementation by only NMOS requires series/parallel connection of multiple NMOS, then in design with CMOS, there will be the exact same number of PMOS in parallel/series connection accordingly. This is called conduction complement.
4. The network made up of PMOSFETs within the CMOS is called 'Pull up Network'. While the one made of NMOSFETs is called 'Pull down Network'.
5. Pull up network raises the output voltage to a high value, while the pull down network does the opposite.

Examples of CMOS Logic:

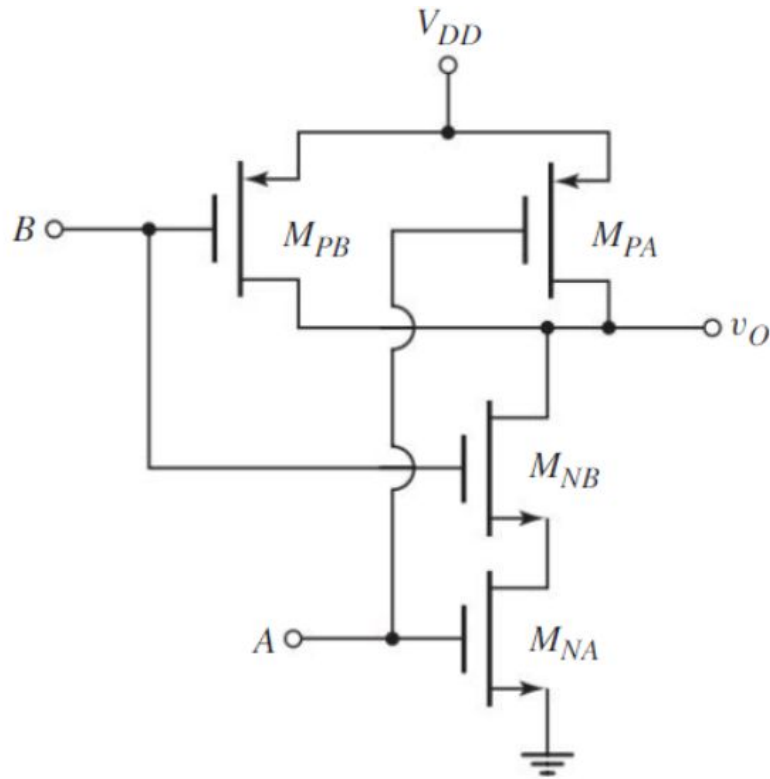


(a)

A	B	v_O
0	0	V_{DD}
V_{DD}	0	0
0	V_{DD}	0
V_{DD}	V_{DD}	0

(b)

Figure 16.37 (a) Two-input CMOS NOR logic circuit and (b) truth table



(a)

A	B	v_O
0	0	V_{DD}
V_{DD}	0	V_{DD}
0	V_{DD}	V_{DD}
V_{DD}	V_{DD}	0

(b)

Figure 16.38 (a) Two-input CMOS NAND logic circuit and (b) truth table

Example: Implement the logic function $Y = AB + C(D+E)$ using CMOS Logic.

Solution:

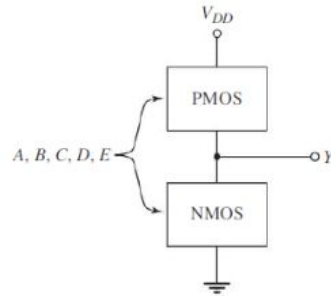


Figure 16.42 General CMOS design

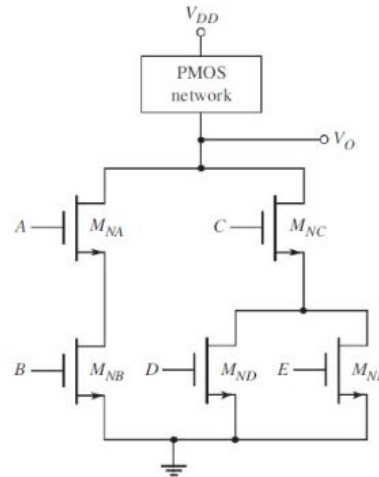


Figure 16.43 NMOS design for Example 16.12

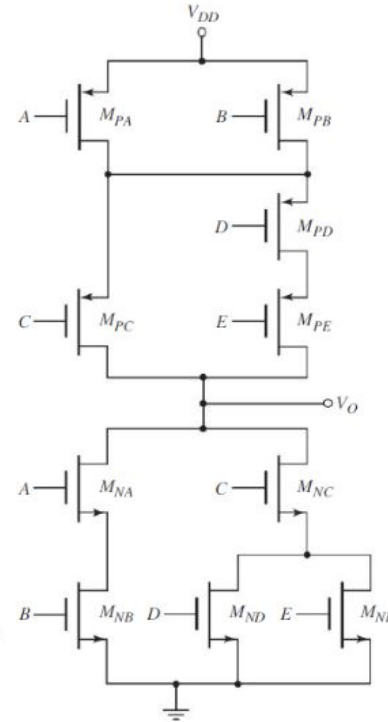


Figure 16.44 Complete CMOS design for Example 16.12

Note: The NMOS, PMOS & CMOS logic circuits for the previous example are actually producing the complement of the desired function. So, to complete the design, we must add an inverter to get the actual function.